

HW Class 06: R Functions

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Overview

This report contains a function that analyzes protein PDB data and plots the B-factors for a specified protein structure.

Questions

Q1: What type of object is returned from the `read.pdb()` function?

This returns a PDB object, which is a list that contains information about the protein structure such as atoms, coordinates, and related structural data.

Q2: What does the `trim.pdb()` function do?

The `trim.pdb()` function keeps only selected parts of a PDB structure like certain chains or atom types and removes the rest.

Q3: What input parameter would turn off the marginal black and grey rectangles in the plots and what do they represent in this case?

To turn off the marginal black and grey rectangles, use `sse = NULL`, or leaving out the `sse` argument. This would represent the secondary structure elements of the protein, such as alpha helices and beta sheets.

Q4. What would be a better plot to compare across the different proteins?

A better plot would be a single combined plot with all three B- Factor trends shown together so the proteins can be compared directly.

Q5. Which proteins are more similar to each other in their B-factor trends. How could you quantify this?

1AKE and 4AKE are more similar to each other in their B-factor trends. This can be quantified by combining the B-factor values with `rbind()`, calculating distances using `dist()`, and using `hclust()`, to create a dendrogram. The two proteins that cluster closest together are the most similar.

Function documentation

Inputs

The function takes one input called `pdb_id`, which is the PDB ID of a protein structure.

What the function does

The function reads in the protein structure from the PDB, trims it to chain A and alpha carbons only, extracts the B-factors, and plots the B-factor values.

Output

The function outputs a B-factor plot for the selected protein.

Code

```
library(bio3d)

plot_protein_bfactor <- function(pdb_id) {

  # Read in the PDB structure
  protein <- read.pdb(pdb_id)

  # Keep only chain A and alpha carbons
  chainA <- trim.pdb(protein, chain = "A", eley = "CA")

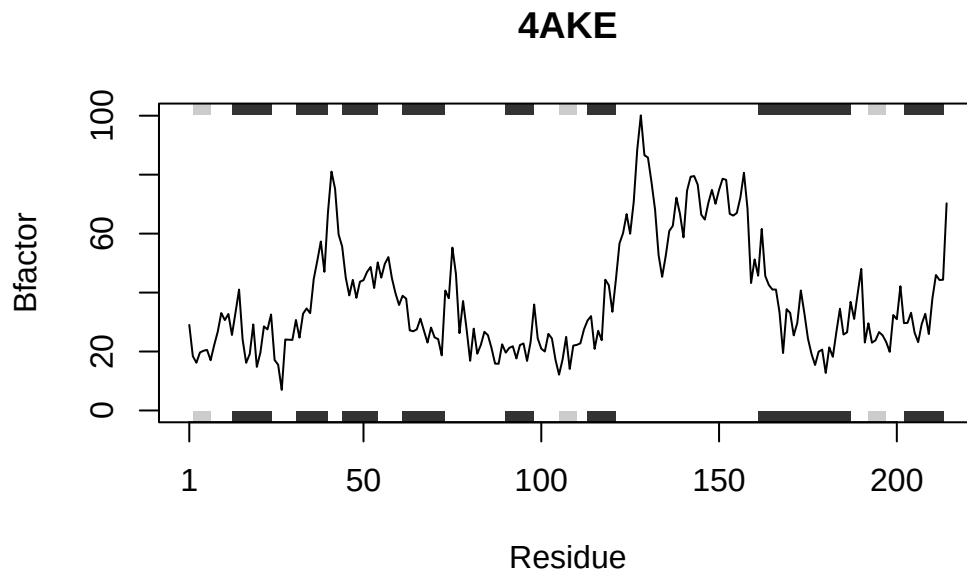
  # Extract B-factors
  b_factors <- chainA$atom$b

  # Plot B-factors
  plot3(b_factors, sse = chainA, typ = "l", ylab = "Bfactor", main = pdb_id)}
```

Example Outputs

```
plot_protein_bfactor("4AKE")
```

Note: Accessing on-line PDB file

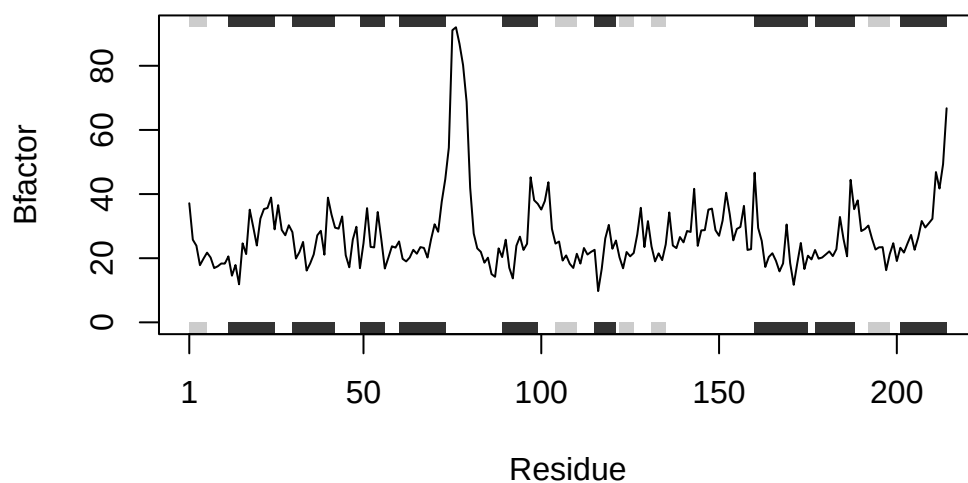


```
plot_protein_bfactor("1AKE")
```

Note: Accessing on-line PDB file

PDB has ALT records, taking A only, rm.alt=TRUE

1AKE



```
plot_protein_bfactor("1E4Y")
```

Note: Accessing on-line PDB file

1E4Y

